

Institut for Mekanik og Produktion

Aarhus Universitet

Øvelser

Forfatter	Noah Rahbek Bigum Hansen	202405538
Kursus	Kompositmaterialer	29016PU022
Semester	Semester 5	



Indhold

Forelæsning 4: Mikromekanik, Pt. I	1
Opgave 4.2	1
(a)	1
(b)	1
Opgave 4.3	3
(a)	3
(b)	3

Forelæsning 4: Mikromekanik, Pt. I

3. September 2026

Opgave 4.2

- (a) Beregn stivheden i 1-retningen for et unidirektionelt glas/epoxy lamina med en fibervolumenfraktion på 0,7 vha. materialeegenskaberne fra tabellen.
- (b) Hvor meget ville stivheden stige hvis kulfiber blev brugt i stedet for glas?

Typical Properties of Fibers (SI System of Units)

Property	Units	Graphite	Glass	Aramid
Axial modulus	GPa	230	85	124
Transverse modulus	GPa	22	85	8
Axial Poisson's ratio	—	0.30	0.20	0.36
Transverse Poisson's ratio	—	0.35	0.20	0.37
Axial shear modulus	GPa	22	35.42	3
Axial coefficient of thermal expansion	$\mu\text{m}/\text{m}/^\circ\text{C}$	-1.3	5	-5.0
Transverse coefficient of thermal expansion	$\mu\text{m}/\text{m}/^\circ\text{C}$	7.0	5	4.1
Axial tensile strength	MPa	2067	1550	1379
Axial compressive strength	MPa	1999	1550	276
Transverse tensile strength	MPa	77	1550	7
Transverse compressive strength	MPa	42	1550	7
Shear strength	MPa	36	35	21
Specific gravity	—	1.8	2.5	1.4

Typical Properties of Matrices (SI System of Units)

Property	Units	Epoxy	Aluminum	Polyamide
Axial modulus	GPa	3.4	71	3.5
Transverse modulus	GPa	3.4	71	3.5
Axial Poisson's ratio	—	0.30	0.30	0.35
Transverse Poisson's ratio	—	0.30	0.30	0.35
Axial shear modulus	GPa	1.308	27	1.3
Coefficient of thermal expansion	$\mu\text{m}/\text{m}/^\circ\text{C}$	63	23	90
Coefficient of moisture expansion	$\text{m}/\text{m}/\text{kg}/\text{kg}$	0.33	0.00	0.33
Axial tensile strength	MPa	72	276	54
Axial compressive strength	MPa	102	276	108
Transverse tensile strength	MPa	72	276	54
Transverse compressive strength	MPa	102	276	108
Shear strength	MPa	34	138	54
Specific gravity	—	1.2	2.7	1.2

Udledning

- (a) Vi bruger rule of mixture, for glasset først:

$$E_{1g} = 85 \text{ GPa} \cdot 0,7 + 3,4 \text{ GPa} \cdot 0,3 = 60,52 \text{ GPa}.$$

Resultat

Stivheden af glaskompositten i 1-retningen er:

$$E_{1g} = 60,52 \text{ GPa}.$$

- (b) Og dernæst for kulfiberkompositten:

$$E_{1c} = 230 \text{ GPa} \cdot 0,7 + 3,4 \text{ GPa} \cdot 0,3 = 162 \text{ GPa}.$$

Resultat

Stivheden af kulfiberkompositten i 1-retningen er:

$$E_{1c} = 162 \text{ GPa}.$$

Endeligt resultat – Opgave 1

- (a) Stivheden af glaskompositten i 1-retningen er:

$$E_{1g} = 60,52 \text{ GPa.}$$

- (b) Stivheden af kulfiberkompositten i 1-retningen er:

$$E_{1c} = 162 \text{ GPa.}$$

Opgave 4.3

- (a) Beregn stivheden i 2-retningen for en unidirektionel glas/epoxy lamina med fibervolumenfraktion på 0,7 vha. materialeegenskaberne i tabellen.
- (b) Hvor meget ville stivheden stige, hvis kulfiber brugtes i stedet

Typical Properties of Fibers (SI System of Units)

Property	Units	Graphite	Glass	Aramid
Axial modulus	GPa	230	85	124
Transverse modulus	GPa	22	85	8
Axial Poisson's ratio	—	0.30	0.20	0.36
Transverse Poisson's ratio	—	0.35	0.20	0.37
Axial shear modulus	GPa	22	35.42	3
Axial coefficient of thermal expansion	μm/m/°C	-1.3	5	-5.0
Transverse coefficient of thermal expansion	μm/m/°C	7.0	5	4.1
Axial tensile strength	MPa	2067	1550	1379
Axial compressive strength	MPa	1999	1550	276
Transverse tensile strength	MPa	77	1550	7
Transverse compressive strength	MPa	42	1550	7
Shear strength	MPa	36	35	21
Specific gravity	—	1.8	2.5	1.4

Typical Properties of Matrices (SI System of Units)

Property	Units	Epoxy	Aluminum	Polyamide
Axial modulus	GPa	3.4	71	3.5
Transverse modulus	GPa	3.4	71	3.5
Axial Poisson's ratio	—	0.30	0.30	0.35
Transverse Poisson's ratio	—	0.30	0.30	0.35
Axial shear modulus	GPa	1.308	27	1.3
Coefficient of thermal expansion	μm/m/°C	63	23	90
Coefficient of moisture expansion	m/m/kg/kg	0.33	0.00	0.33
Axial tensile strength	MPa	72	276	54
Axial compressive strength	MPa	102	276	108
Transverse tensile strength	MPa	72	276	54
Transverse compressive strength	MPa	102	276	108
Shear strength	MPa	34	138	54
Specific gravity	—	1.2	2.7	1.2

Udledning

- (a) Vi bruger her den inverse rule of mixture:

$$E_{2g} = \frac{85 \text{ GPa} \cdot 3,4 \text{ GPa}}{85 \text{ GPa} \cdot 0,3 + 3,4 \text{ GPa} \cdot 0,7} = 10,3659 \text{ GPa.}$$

Resultat

Stivheden af glaskompositten i 2-retningen er:

$$E_{1g} = 10,37 \text{ GPa.}$$

- (b) Hvis kulfiber brugtes i stedet:

$$E_{2c} = \frac{230 \text{ GPa} \cdot 3,4 \text{ GPa}}{230 \text{ GPa} \cdot 0,3 + 3,4 \text{ GPa} \cdot 0,7} = 10,9554 \text{ GPa.}$$

Resultat

Stivheden af kulfiberkompositten i 2-retningen er:

$$E_{1c} = 10,96 \text{ GPa.}$$

Endeligt resultat – Opgave 2

- (a) Stivheden af glaskompositten i 2-retningen er:

$$E_{1g} = 10,37 \text{ GPa.}$$

- (b) Stivheden af kulfiberkompositten i 2-retningen er:

$$E_{1c} = 10,96 \text{ GPa.}$$